

EC3008 — Financial Strategy

Winter 2025 Examination · Paper 1 · Full Marking Scheme

Module	EC3008 — Financial Strategy
Internal Examiner	Dr Ryan
Institution	University College Cork
Session	Winter 2025, Paper 1
Duration	1.5 hours
Structure	Section A — Short Questions: attempt 2 of 4 (30% of the paper). Section B — Long Questions: attempt 2 of 2 , i.e. both are compulsory (70% of the paper). Use separate answer books for each section.
Materials	None specified

Coverage note. This scheme answers **all four** Section A questions in full, since you'll choose your best two under exam conditions, plus both compulsory Section B questions.

Source basis. Content follows Dr Ryan's own EC3008 lecture material (Modigliani-Miller, Trade-Off Theory, Pecking Order Theory, the Microsoft/Activision case study) and the module's assigned readings, including the in-class November 2023 guest talk by Anna-Marie Turley (Enterprise Ireland) and her co-authored paper on EI supports for female entrepreneurs (Turley et al., 2024), and Cumming, Leboeuf & Schwienbacher's (2020) paper on crowdfunding funding models, which underlies the Lynn and Rosati (2021) question on this paper.

Flagged: facts verified beyond the course pack. Four sub-questions on this paper (A1, A2, B1(c)'s KDB/EBITDA detail) turn on specific, fast-moving figures — Ireland's corporation tax rate structure, the Section 110 regime's technical requirements, the Knowledge Development Box's current effective rate, and the

interest limitation rule's threshold — that were not set out in the uploaded lecture material. These were verified against current Irish Revenue and Department of Finance guidance rather than assumed, and are marked with a checknote at the relevant question. If Dr Ryan's own lecture slides state a different figure (e.g. an older KDB rate), defer to the lecture material for the exam itself.

Section A

Short Questions — attempt 2 of 4 — 30% of the paper — 100 marks per question answered

A1

[100 marks]

Using examples, explain why Ireland's tax rate makes debt less tax-efficient for companies compared to countries with higher tax rates.

FULL MODEL ANSWER

The mechanism linking tax rate to debt's value: the entire tax advantage of debt comes from the **interest tax shield** — interest payments are deductible against taxable profit, so every euro of interest paid reduces the tax bill by (interest × corporate tax rate). That formula makes the size of the benefit **directly proportional to the tax rate**: the same euro of interest is worth less as a tax shield in a low-tax country than in a high-tax one, because there is less tax to shield the income from in the first place.

Jurisdiction (illustrative)	Corporate tax rate	Tax shield on €1,000,000 interest
Ireland (trading income)	12.5%	€125,000
A ~25% jurisdiction (e.g. UK main rate, or typical US combined federal+state)	25%	€250,000

Result: identical debt, identical interest cost — but the after-tax benefit of carrying that debt is **half as large** for an Irish company as for an equivalent company in a 25% jurisdiction. Since the Trade-Off Theory (see Q. B1(c)) says firms lever up until the marginal tax-shield benefit of one more euro of debt equals the marginal cost of financial distress it creates, a smaller tax-shield benefit at every debt level pushes the theoretical optimal debt level **lower** for an Irish company than for an otherwise identical company facing a higher tax rate — the marginal benefit curve is flatter, so it is crossed by the marginal distress-cost curve sooner, at a lower level of debt.

EXAM TECHNIQUE

Ireland's **12.5%** rate applies to ordinary trading income; a **25%** global minimum

effective rate under OECD Pillar Two applies only to large multinational groups with consolidated revenue above €750 million — the vast majority of Irish companies are unaffected by Pillar Two and pay 12.5%. This distinction matters for a complete answer: don't imply Ireland's rate has risen to 15% or 25% generally — for most companies it hasn't. These rate figures were verified against current Revenue/Department of Finance guidance since they weren't set out in the lecture material — check them against Dr Ryan's own slides if a different figure was given in class.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
76-100	76-100	Correctly states the interest tax shield formula (interest × tax rate) AND a correct, quantified example comparing Ireland's 12.5% rate to a higher rate AND explicitly links this to a lower optimal debt level via Trade-Off Theory reasoning.
51-75	51-75	Correct qualitative explanation that a lower tax rate means a smaller tax shield, with an example, but no explicit link to the optimal-debt-level implication.
26-50	26-50	States that debt is "less tax efficient" in Ireland without correctly explaining the interest-tax-shield mechanism, or the example given doesn't correctly apply a tax rate to an interest figure.
0-25	0-25	No credible mechanism given, or confuses corporate tax rate with an unrelated concept (e.g. VAT, personal income tax).

A2

[100 marks]

A Section 110 company is a unique Irish SPV structure. Describe its key features and explain how it is used in securitization transactions in Ireland.

FULL MODEL ANSWER

What a Section 110 company is: an Irish tax-resident special purpose vehicle (SPV) that qualifies for the regime set out in **Section 110 of the Taxes Consolidation Act 1997**. Where the conditions are met, the company is taxed broadly as an ordinary trading company but structured so that its taxable profit is reduced close to zero — achieving what practitioners call **tax neutrality**, rather than statutory tax exemption.

Key features

- **Irish tax residence and arm's-length dealing:** the company must be Irish tax resident, and (aside from the payment of profit-participating interest itself) must conduct its transactions on an arm's-length, commercial basis.
- **Holds "qualifying assets":** broadly defined financial assets — loans, receivables, bonds, derivatives and securities — extended by the Finance Act

2011 to also cover plant and machinery, leased assets, commodities and aircraft, which is why Section 110 vehicles are heavily used in aircraft leasing as well as securitisation.

- **€10 million minimum, on a day-one basis:** the market value of qualifying assets acquired must be at least €10 million on the date of acquisition — a threshold that applies only at that initial point, not on an ongoing basis.
- **Tax neutrality achieved via profit-participating notes (PPNs):** the SPV funds itself by issuing notes whose interest is contractually tied to the SPV's own profit. Because that interest is deductible, it can be structured to exactly offset the SPV's taxable income, leaving little or no Irish tax actually payable — the SPV is not tax-exempt by statute, it is tax-neutral by structure.
- **Withholding tax and stamp duty relief:** qualifying interest and dividend payments can be exempt from Irish withholding tax, and the issue and transfer of the SPV's notes is exempt from stamp duty — both of which matter directly for cross-border securitisation where investors sit outside Ireland.
- **Notification to Revenue:** the SPV must file a Form S110 within eight weeks of first acquiring qualifying assets in order to avail of the regime at all.

Use in securitisation transactions

The mechanism: an originating bank or lender (the **originator**) sells a pool of assets — mortgages, loans, receivables — to the Section 110 SPV. The SPV funds that purchase by issuing asset-backed notes to investors. Because the SPV's interest on those notes is structured as profit-participating, the interest income the SPV earns from the underlying loan pool flows through to noteholders largely untaxed at the SPV level, and the withholding-tax and stamp-duty exemptions mean the cross-border flow of interest to international investors isn't eroded by Irish tax at the vehicle stage. This is precisely the same **originate-to-distribute** structure discussed for shadow banking more generally (see the EC3005 Winter 2025 scheme, Q. A2(a)) — a Section 110 company is the specific Irish tax-law vehicle that implements it.

EXAM TECHNIQUE

Section 110's technical requirements (the €10m day-one threshold, Form S110, the specific asset categories, and the current withholding/stamp duty reliefs) were verified against Revenue's own guidance and the Taxes Consolidation Act 1997, since the course pack's coverage of Section 110 was limited to its role in shadow banking and multinational tax planning rather than the structural detail. Cross-check against Dr Ryan's own slides for the depth of technical detail actually expected in the exam answer.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
76-100	76-100	Correctly defines a Section 110 company as an Irish tax-resident qualifying-company regime AND names at least three specific key features (qualifying assets, €10m threshold, PPN-based tax neutrality, withholding/stamp duty relief) AND correctly explains the originator → SPV → noteholder securitisation mechanism.
51-75	51-75	Correctly identifies Section 110 as an Irish SPV tax regime with at least two features named, and gives a broadly correct but under-detailed account of its role in securitisation.
26-50	26-50	Vague reference to Section 110 as a "tax-efficient vehicle" without naming specific features or explaining the securitisation mechanism.
0-25	0-25	No credible description given, or Section 110 is confused with an unrelated structure.

A3**[100 marks]**

Different stakeholders prioritize different metrics including ROE, WACC, P/E, EVA, ROCE. From these five metrics, identify which one matters most to (i) equity investors and (ii) management. Explain why their priorities differ.

FULL MODEL ANSWER

(i) Equity investors — Return on Equity (ROE): ROE (net income ÷ shareholders' equity) measures the return generated specifically on the capital that **shareholders themselves** have put at risk — it is, definitionally, the metric closest to "what am I, the equity holder, actually earning on my investment." P/E is also equity-investor-relevant, but it's a *valuation* multiple (what the market is willing to pay for a euro of earnings) rather than a measure of the underlying business's own performance — ROE speaks to whether the company is generating good returns in the first place, which is what ultimately drives the earnings that P/E is a multiple of.

(ii) Management — Weighted Average Cost of Capital (WACC) and Economic Value Added (EVA), taken together: management's job is capital allocation — deciding which projects to fund — and that decision requires a hurdle rate, which is exactly what WACC provides: the blended cost of every source of capital (debt and equity) the firm uses. A project only creates value if its return exceeds WACC. **EVA** operationalises this directly into a single performance metric — $EVA = NOPAT - (WACC \times \text{Invested Capital})$ — which is precisely why EVA (built on top of WACC) is the metric most directly used internally to assess whether management's capital allocation decisions actually created value, rather than merely grew accounting profit.

Why the priorities differ: the difference comes down to **what each party controls and is accountable for**. An equity investor doesn't control the firm's day-to-day capital allocation — they observe the outcome and price it, so ROE (an outcome measure on their specific capital) and P/E (how the market prices that outcome) are what's directly relevant to them. Management *does* control capital allocation and is accountable for using both debt and equity capital efficiently, not just equity — so their relevant benchmark has to price the full blended cost of capital (WACC) and translate it into a decision rule (EVA) that can be applied project-by-project, not just observed at the aggregate, whole-company level the way ROE is.

EXAM TECHNIQUE

ROCE (return on capital employed) is a reasonable alternative answer for management's metric instead of WACC/EVA — it also captures returns on the full capital base rather than just equity. What earns marks is correctly reasoning from *what each stakeholder controls* to *which metric matches that scope*, not memorising a fixed pairing.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
76-100	76-100	Correct metric identified for each stakeholder AND each metric's definition/formula correctly given AND the "why" is grounded in what each stakeholder controls or is accountable for, not just asserted.
51-75	51-75	Correct metrics identified with correct definitions, but the explanation of why priorities differ is generic ("investors care about returns, managers care about the business") rather than reasoned from control/accountability.
26-50	26-50	One metric correctly matched to its stakeholder; the other is incorrect or undefended.
0-25	0-25	Neither metric convincingly matched to its stakeholder, or the five metrics are not meaningfully distinguished from one another.

A4

[100 marks]

Using examples, distinguish between aggressive and conservative capital structures.

FULL MODEL ANSWER

The core distinction: an **aggressive capital structure** relies heavily on debt — particularly short-term debt — relative to equity, maximising leverage (and the tax shield) at the cost of higher financial risk. A **conservative capital structure** relies predominantly on equity and long-term financing, minimising leverage and financial risk at the cost of forgoing some of the tax shield and potentially diluting returns to existing shareholders less aggressively than debt would allow.

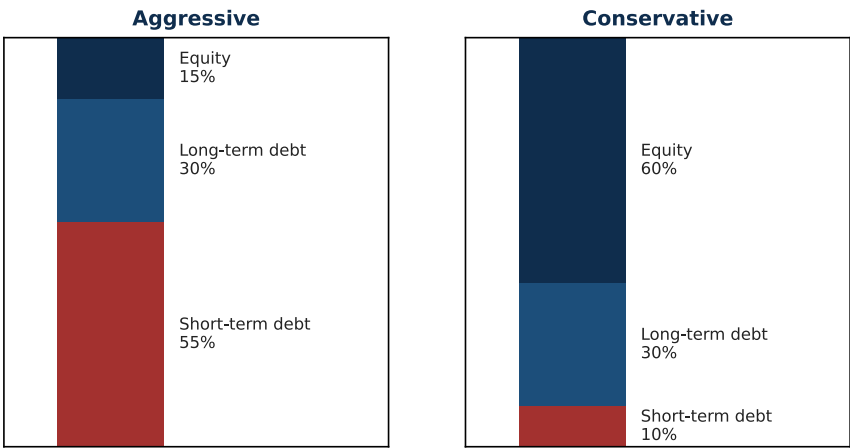


Fig. 1 — An aggressive structure funds most of the balance sheet with short-term and long-term debt; a conservative structure funds most of it with equity, matching the pattern discussed for Ryanair's net-cash position under Trade-Off Theory (see Q. B1(c)).

	Aggressive	Conservative
Financing mix	Debt-heavy, tilted toward short-term debt	Equity-heavy, long-term financing
Financial risk	Higher — mandatory interest/principal payments regardless of performance	Lower — dividends are discretionary, not contractual
Cost of capital	Lower WACC (debt is cheaper than equity) if distress risk stays contained	Higher WACC, but more resilient to downturns
Financial flexibility	Lower — debt covenants and repayment obligations constrain future choices	Higher — spare debt capacity available if opportunities arise
Example	A leveraged buyout target loaded with acquisition debt to boost equity returns	Apple Inc. historically — large cash reserves, minimal debt relative to its size

Why firms choose one over the other: the choice reflects the same trade-off underlying both Trade-Off Theory and Pecking Order Theory (see Q. B1) — an aggressive structure captures more of the tax shield and can amplify shareholder returns when things go well, but raises the probability and cost of financial distress; a conservative structure sacrifices some of that upside to preserve flexibility and reduce the risk of default, which matters more to firms in volatile industries, firms with valuable growth options they don't want debt covenants to constrain, or firms (like Ryanair, historically) exposed to large unpredictable cost shocks.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
76-100	76-100	Correct distinction stated in terms of leverage and debt maturity mix AND at least one concrete, correctly-reasoned example of each AND the risk/flexibility trade-off underlying the choice is explained.
51-75	51-75	Correct distinction with at least one example, but the underlying risk/flexibility trade-off is only asserted, not explained.
26-50	26-50	Vague distinction ("aggressive uses more debt") without a worked example or explanation of the trade-off.
0-25	0-25	No meaningful distinction drawn, or capital structure is confused with an unrelated concept (e.g. working capital policy).

Section B · Question 1

Capital structure theory applied to an Irish tech company — 100 marks — compulsory

B1 — scenario

[marks]

Consider a successful Irish tech company that has recently gone public and is evaluating its capital structure strategy.

B1(a)

[33 marks]

State and explain the two propositions of the Modigliani-Miller Theorem. List four market imperfections that make capital structure matter.

B1(b)

[33 marks]

Using Pecking Order Theory, explain why Irish technology companies typically maintain conservative debt-to-equity ratios despite possessing valuable intellectual property. Your answer should address the financing hierarchy and underwriting costs.

B1(c)

[34 marks]

Apply the Trade-Off Theory to determine the optimal debt level for this Irish tech company. In your answer define and explain (i) KDB (Knowledge Development Box) and how it affects the value of debt tax shields for software patents, and (ii) EBITDA and how its rule restricts debt usage.

FULL MODEL ANSWER

Proposition I: in a perfect capital market (no taxes, no bankruptcy costs, symmetric information, no agency costs), **firm value is independent of capital structure:** $V(\text{levered}) = V(\text{unlevered})$. Any attempt to change the debt/equity mix merely redistributes the same total value and risk between claimholders rather than creating or destroying value. This is enforced by **arbitrage** — if two otherwise-identical firms were valued differently purely because of leverage, investors could use "home-made leverage" (borrowing or lending on their own account to replicate any capital structure they want) to profit from the discrepancy until it closed.

Proposition II: the **cost of equity rises linearly with leverage:** $r_e = r_0 + (r_0 - r_d) \times D/E$, where r_0 is the all-equity cost of capital and r_d is the cost of debt. As a firm takes on more debt, debtholders' prior claim on cash flows makes the residual claim (equity) riskier, so shareholders demand a higher return to compensate — meaning debt is cheaper than equity, *but* more debt makes the remaining equity correspondingly more expensive, which is precisely why Proposition I holds: the two effects offset exactly, leaving the firm's overall (blended) cost of capital and value unchanged.

MM Proposition II: Cost of Equity Rises with Leverage

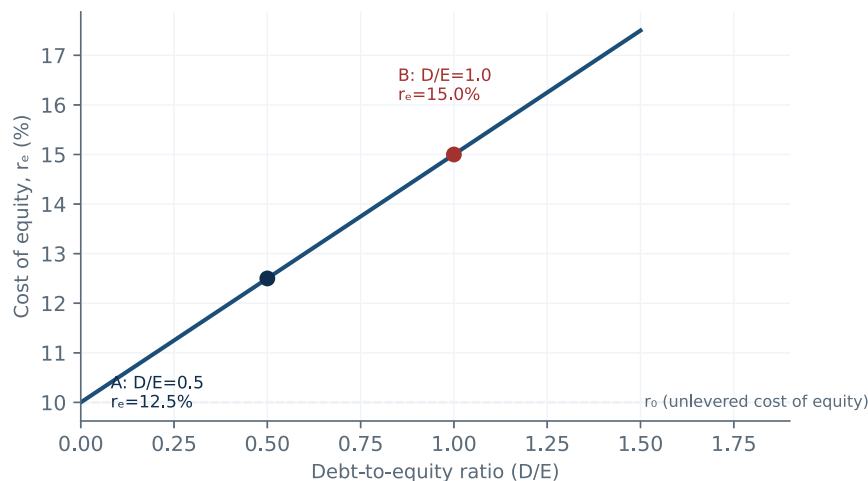


Fig. 1 — MM Proposition II: with $r_0=10\%$ and $r_d=5\%$, the cost of equity rises from 12.5% at $D/E=0.5$ to 15% at $D/E=1.0$ — the cost of the extra leverage is fully absorbed by shareholders demanding a higher return.

Four market imperfections that make capital structure matter

- **Taxes:** interest is tax-deductible, dividends are not — creating a tax shield that makes debt cheaper on an after-tax basis and giving firms a reason to prefer some debt (see Q. A1 for how the size of this effect depends directly on the tax rate).
- **Bankruptcy (financial distress) costs:** direct costs (legal, administrative) and indirect costs (lost sales, higher supplier costs, lost key staff) of financial distress rise with leverage, discouraging firms from pushing debt too far.
- **Information asymmetry:** managers know more about the firm's true prospects than outside investors, so financing choices are read as signals — issuing equity, in particular, can be read as a signal that management believes the stock is overvalued.
- **Agency costs:** conflicts of interest between shareholders, managers and debtholders — e.g. shareholders' incentive to risk-shift into riskier projects once debt is in place, or managers' incentive to over-retain free cash flow — mean the capital structure choice also has to manage who is incentivised to do what.

EXAM TECHNIQUE

Transaction costs and market timing are also legitimate fifth/sixth imperfections covered in the lecture material — if you list one of these instead of agency costs or information asymmetry, that's equally creditable. The question asks for four, not a specific four.

Mark Allocation — Part (a) (33 marks)

Tier	Marks	What separates this tier
25-33	25-33	Both propositions correctly stated with their formulas AND the arbitrage enforcement mechanism for Proposition I explained AND four correctly named and briefly explained market imperfections.
17-24	17-24	Both propositions stated correctly but without the arbitrage mechanism, and at least three imperfections named.
9-16	9-16	Only one proposition correctly stated, or imperfections listed with no explanation of how each makes capital structure matter.
0-8	0-8	Neither proposition correctly stated.

The pecking order and why it applies with extra force to tech

IP: Pecking Order Theory (Myers and Majluf, 1984) ranks financing sources by the **information asymmetry** — and therefore the "lemon premium" — an

outside investor demands: internal funds first (no asymmetry), debt second (lower asymmetry, senior claim), equity last (highest asymmetry, residual claim, and a negative signal since issuing equity can be read as management believing the stock is overvalued).

Pecking Order Theory: The Financing Hierarchy

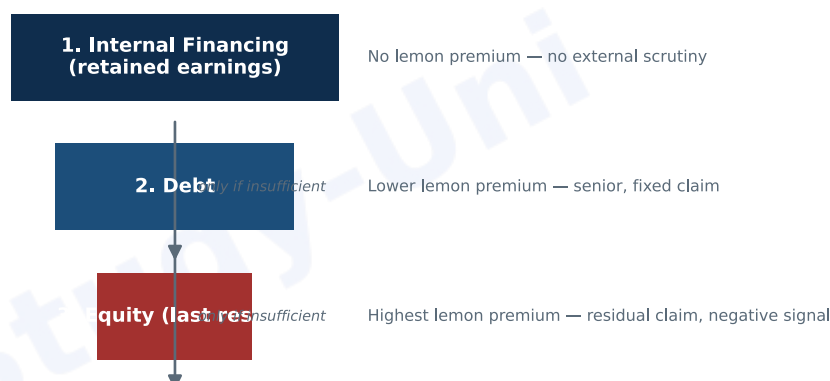


Fig. 2 — The financing hierarchy: firms exhaust internal funds before turning to debt, and turn to equity only as a last resort, since the lemon premium — and therefore the cost of external financing — rises at each step down.

Why this pushes Irish tech companies toward conservative D/E

specifically: the information asymmetry that drives the whole theory is **unusually severe** for an IP-heavy tech company. Software patents and codebases are hard for outside lenders to value or use as collateral — a bank cannot easily repossess and resell a patent portfolio the way it could a factory, so **debt against IP-heavy assets is itself expensive or unavailable** even though debt would normally be pecking order's preferred external source. Combined with the fact that a profitable, well-funded tech company (post-IPO, with cash reserves) has less *need* for external financing of either kind, the pecking order prediction that "profitable firms with internal funds borrow less" applies especially strongly — this is exactly the Apple Inc. pattern noted in the lecture material (highly profitable, large cash reserves, minimal debt relative to size).

Underwriting costs: issuing external capital — whether debt or equity — carries real transaction costs (underwriting fees, legal and regulatory costs) that are part of Pecking Order Theory's rationale for preferring internal funds outright. The Facebook 2012 IPO paid underwriting fees of roughly 1.1% of the \$16 billion raised (about \$176 million) — low for the sector because of the deal's scale and profile — while typical smaller offerings pay 3–7%. For a smaller, less high-profile Irish tech company, underwriting costs on a public debt or equity issue would sit far closer to that higher end, making external financing of any kind disproportionately expensive relative to a large-cap issuer — reinforcing the preference for internal funds and, where external financing is unavoidable, for smaller-scale, less costly forms of debt over full public offerings.

Mark Allocation — Part (b) (33 marks)

Tier	Marks	What separates this tier
25-33	25-33	Correctly states the financing hierarchy and the information-asymmetry/lemon-premium rationale AND explains why IP-heavy assets are specifically hard to collateralise, sharpening the pecking order prediction for tech firms AND discusses underwriting costs as a distinct, correctly-explained driver.
17-24	17-24	Correctly states the financing hierarchy and applies it to explain

conservative D/E, but the IP-collateral point or the underwriting-cost discussion is missing or only asserted.

9-16	9-16	States the pecking order hierarchy correctly but does not apply it specifically to Irish tech companies or IP.
0-8	0-8	Pecking Order Theory not correctly described, or conflated with Trade-Off Theory.

Trade-Off Theory applied to this company: Trade-Off Theory says the firm should increase debt only up to the point where the **marginal tax-shield benefit** of another euro of debt equals the **marginal expected cost of financial distress** it creates — the optimal debt level sits at the peak of firm value against debt, not at zero debt and not at maximum debt.

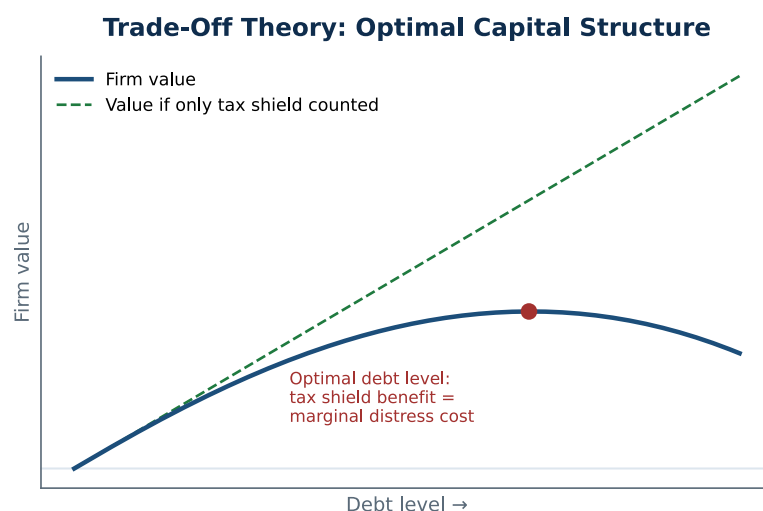


Fig. 3 — Firm value rises with debt as the tax shield accumulates, then falls as financial distress costs grow faster than the shield — the optimal debt level is where the two forces exactly offset at the margin.

(i) KDB and its effect on debt tax shields for software patents

Definition: the **Knowledge Development Box** is an OECD-compliant Irish tax regime (Finance Act 2015) that applies a reduced effective corporation tax rate to profits from qualifying IP — including patented inventions and copyrighted software — developed through R&D carried out in Ireland, under the OECD's modified nexus approach linking the relief to where the R&D actually happened.

Basis of interest deduction	Tax rate applied	Value of tax shield on €1,000,000 interest
Interest offset against ordinary trading profit	12.5% (standard rate)	€125,000
Interest offset against KDB-qualifying software IP profit	10% (KDB effective rate)	€100,000

The effect on debt tax shields: the value of any tax deduction is the deduction multiplied by the tax rate it's set against — so where a software company's profit is **KDB-qualifying** (taxed at an effective 10% rather than the standard 12.5%), the same euro of interest expense shields *less* tax, exactly as a lower national tax rate does more broadly (see Q. A1). For an IP-heavy tech company with a meaningful share of profit inside the KDB, this **further weakens** the tax-shield argument for debt specifically on top of Ireland's already-low 12.5% headline rate — reinforcing, from the tax side, the same conservative-leverage conclusion Pecking Order Theory reached in Part (b) from the information-asymmetry side.

(ii) EBITDA and the interest limitation rule

Definition: Ireland's **Interest Limitation Rule** (Part 35D TCA 1997, transposing Article 4 of the EU Anti-Tax-Avoidance Directive, effective for accounting periods from 1 January 2022) caps the net interest a company can deduct for tax purposes at **30% of tax-adjusted EBITDA** (earnings before interest, tax, depreciation and amortisation), or €3 million if higher — with interest above that cap not lost outright but carried forward for deduction in future periods once there is spare capacity.

Metric	Value
Tax-EBITDA (illustrative)	€10,000,000
30% of EBITDA	€3,000,000
€3m de minimis alternative	€3,000,000
Maximum deductible net interest (higher of the two)	€3,000,000

How it restricts debt usage: the rule directly caps the tax benefit available from leverage once net interest exceeds 30% of EBITDA, regardless of how much debt the company could otherwise service — pushing the trade-off curve's peak **leftward, toward a lower debt level**, since interest beyond the cap earns no immediate tax shield at all. For a growth-stage tech company with relatively modest current EBITDA relative to its ambitions, this cap can bind well before financial distress risk alone would have discouraged further borrowing — a second, independent reason (alongside the KDB point above) that the model in Part (c) predicts a conservative optimal debt level for this company.

EXAM TECHNIQUE

The KDB effective rate (10%, increased from 6.25% for accounting periods from 1 October 2023) and the interest limitation rule's 30%/€3m thresholds were verified against current Revenue guidance, since neither appeared in the uploaded course pack. If Dr Ryan's own slides give the pre-2023 KDB rate (6.25%) or a different EBITDA threshold, use the figure taught in lectures for the exam itself — the mechanism and direction of the argument are unaffected either way.

Mark Allocation — Part (c) (34 marks)

Tier	Marks	What separates this tier
26-34	26-34	Trade-Off Theory's optimal-debt-level logic correctly applied AND KDB correctly defined with its effect on the tax shield's value correctly reasoned AND the EBITDA interest limitation rule correctly defined with its restriction on debt usage correctly reasoned.
17-25	17-25	Trade-off logic and one of KDB/EBITDA correctly explained; the other is defined but its effect on the optimal debt level is not reasoned through.
9-16	9-16	KDB and EBITDA rule both named but only vaguely described, without applying either to the optimal debt level question asked.
0-8	0-8	Neither KDB nor the EBITDA rule correctly defined.

Question B1 Total: 100 Marks

Section B · Question 2

Enterprise Ireland funding, convertible bonds, crowdfunding — 100 marks — compulsory

B2(a)

[33 marks]

Drawing from the insights shared by Anna-Marie Turley, Head of Fintech at Enterprise Ireland, analyze **two** distinct funding options Enterprise Ireland provides to support start-up businesses in Ireland.

B2(b)

[33 marks]

Define convertible bonds and discuss how the different accounting treatment of these instruments under IFRS versus US GAAP influences global capital market structures. Your answer should cover the impact on leverage ratios, reported profitability and credit ratings.

B2(c)

[34 marks]

According to Lynn and Rosati (2021), crowdfunding and token offerings are transforming entrepreneurial finance. Explain how crowdfunding addresses the information asymmetry problem differently than traditional venture capital. In your answer: (i) define "all-or-nothing" versus "keep-it-all" funding models; (ii) discuss one advantage and one limitation of crowdfunding for Irish SMEs.

FULL MODEL ANSWER

Framing: Enterprise Ireland (EI) is the Irish government agency responsible for supporting the development of Irish-owned businesses, and runs several distinct, non-mutually-exclusive equity funding programmes aimed at different stages of a start-up's growth — precisely the kind of staged funding landscape Anna-Marie Turley's own work on EI supports for entrepreneurs documents.

Programme	Stage / purpose	Structure
High Potential Start-Up (HPSU)	EI's most consistent programme — for start-ups with the potential for significant job creation and export sales	Equity investment, sized to the business's funding needs and growth stage
Competitive Start Fund (CSF)	Earlier-stage seed funding, awarded competitively	€50,000 equity investment (discontinued after 2022, per EI's own published programme data)
Pre-Seed Start Fund (PSSF)	Newer early-stage programme introduced to fill the gap left by CSF	€50,000 or €100,000 equity investment

Two funding avenues analysed — HPSU and the Competitive Start

Fund/Pre-Seed Start Fund: the **HPSU programme** is EI's flagship equity vehicle: rather than a fixed grant amount, it sizes an equity investment to a start-up judged to have genuine potential for significant job creation and export sales, and — as EI's own programme data shows — has been the most consistently used of EI's funding streams over time. The **CSF/PSSF family**, by contrast, targets the earlier, pre-revenue or pre-traction stage with smaller, standardised equity cheques (€50,000, or up to €100,000 under PSSF) awarded through a competitive process — designed to get a larger number of very early-stage companies to the point where they have enough traction to credibly seek HPSU-level or private investment. The two are complementary rather than substitutes: a company might plausibly raise CSF or PSSF funding first, then graduate to HPSU support once it has demonstrated the growth and export potential HPSU specifically looks for.

EXAM TECHNIQUE

EI's programme names and structures (HPSU, CSF, PSSF and their funding amounts) are drawn directly from the module's own assigned reading — Turley, Ryan and Doyle's paper on EI supports for female entrepreneurs — rather than external sourcing, since this is the most directly relevant and lecturer-endorsed source available for this question. If a different pairing of EI programmes was emphasised in the November 2023 guest talk itself, use that pairing instead — the analysis structure (stage, purpose, funding structure, how they complement each other) transfers regardless of which two programmes you pick.

Mark Allocation — Part (a) (33 marks)

Tier	Marks	What separates this tier
25-33	25-33	Two distinct, correctly-named EI funding programmes analysed AND each correctly described in terms of stage/purpose and funding structure AND how the two relate to each other in a start-up's funding journey is explained.
17-24	17-24	Two EI programmes named and broadly correctly described, but the relationship between them (or their fit within a start-up's funding stage) is not explained.
9-16	9-16	Only one EI programme correctly described in any detail; the second is vague or generic.
0-8	0-8	No specific EI programmes correctly named or described.

Definition: a **convertible bond** is a debt instrument that gives the holder the right — but not the obligation — to convert it into a fixed number of the issuer's common shares, combining a straight bond with an embedded equity call option. Because investors value that conversion right, convertible bonds typically carry a lower coupon than an equivalent straight bond, making them a cheaper source of cash financing for the issuer in exchange for potential future dilution.

The accounting divergence

	IFRS (IAS 32)	US GAAP (post-ASU 2020-06)
Treatment	Bifurcated: split into a debt component and an equity component	Recorded as a single liability at full par value for most standard converts
Initial	Lower — measured as the present value of the bond discounted at the market	Higher — the full par value

liability	rate for equivalent straight (non-convertible) debt	of the instrument
Equity component	The residual (fair value of the whole instrument minus the debt component) sits in shareholders' equity, not remeasured afterward	None recognised — the conversion option is not separately valued for most standard converts
Interest expense over time	Higher — the discounted liability accretes up to par via the effective-interest method, adding an accretion charge on top of the coupon	Lower — interest expense simply reflects the stated coupon rate, no accretion

Impact on leverage ratios: because IFRS records only the discounted debt component as a liability (with the residual sitting in equity), an IFRS reporter's convertible bond shows up as **less debt and more equity** than the same instrument would under US GAAP, which puts the **full par value into debt** with nothing allocated to equity — so a US GAAP filer's debt-to-equity ratio on an otherwise identical convertible issue will appear structurally higher than an IFRS filer's, purely as an accounting artefact rather than a real difference in economic leverage.

Impact on reported profitability: IFRS's accretion mechanism means interest expense rises above the stated coupon over the bond's life, depressing reported net income more than under US GAAP, where interest expense stays at the flat coupon rate — so, all else equal, an **IFRS reporter shows lower reported profitability** on the same convertible issue than a comparable US GAAP filer, even though the underlying cash paid to bondholders is identical.

Impact on credit ratings: credit rating agencies generally look through to the economic substance of a convertible — its debt-like cash obligations and its equity-like conversion option — rather than mechanically adopting either accounting framework's balance sheet presentation, so ratings analysis tends to normalise across IFRS and US GAAP filers. The practical effect is that **raw, un-adjusted leverage or profitability ratios pulled straight from IFRS versus US GAAP filings are not directly comparable** without this normalisation — an analyst comparing an Irish (IFRS) tech company's convertible-adjusted leverage to a US (GAAP) competitor's needs to correct for the accounting difference itself before drawing any conclusion about which company is actually more conservatively financed.

EXAM TECHNIQUE

US GAAP's treatment changed materially with ASU 2020-06 (effective for most companies from fiscal year 2022 onward) — before that, GAAP also bifurcated many convertibles in a more complex way than IFRS still does. If you're citing a pre-2022 source or example, note that the simplification to "single liability, full par value" is the current rule, not the historical one.

Mark Allocation — Part (b) (33 marks)

Tier	Marks	What separates this tier
25-33	25-33	Convertible bonds correctly defined AND the IFRS bifurcation vs. US GAAP single-liability treatment correctly contrasted AND all three named impacts (leverage, profitability, credit ratings) correctly reasoned through from the accounting mechanics, not just asserted.
17-24	17-24	Correct definition and correct IFRS/GAAP contrast, but only one or two of the three named impacts correctly reasoned through.
9-16	9-16	Convertible bonds correctly defined, but the IFRS/GAAP accounting difference is vague or incorrect.

How crowdfunding addresses information asymmetry differently from VC: traditional venture capital solves the information-asymmetry problem through **concentrated, expert due diligence** — a small number of specialist investors each spend heavily on evaluating a single deal, then monitor it closely post-investment (see the delegated-monitoring point in Q. A1(c) of the EC3005 scheme). Crowdfunding instead solves the same problem through **aggregation and market signal** — no single backer needs to conduct expert due diligence, because the willingness of a large number of independent backers to pledge money is itself the signal of quality and demand, cheaply revealing information that would otherwise require costly individual screening. This is why crowdfunding can reach much earlier-stage, smaller, or more numerous ventures than VC's due-diligence-intensive model can economically support.

(i) All-or-nothing vs. keep-it-all funding models

Definitions (Cumming, Leboeuf and Schwienbacher, 2020): in the **"Keep-It-All" (KIA)** model, the entrepreneur keeps the entire amount pledged regardless of whether the stated funding goal is reached, which shifts the risk of an underfunded project onto the backers. In the **"All-or-Nothing" (AON)** model, the entrepreneur receives nothing at all unless the funding goal is met in full, which shifts that risk onto the entrepreneur instead — but functions as a credible commitment signal to backers that the entrepreneur won't attempt an underfunded project, letting AON campaigns set larger goals and, empirically, succeed at meeting their goals more often than KIA campaigns.

(ii) One advantage and one limitation for Irish SMEs

Advantage — access without collateral or trading history: crowdfunding lets an Irish SME raise capital based on market/product demand signals rather than the collateral, trading history, or IP-backed security a bank or VC would typically require (compare Q. B1(b) on how hard IP is to collateralise) — directly widening access to finance for exactly the kind of early-stage, asset-light company that struggles most under the traditional financing hierarchy.

Limitation — small scale and market-size constraints: Ireland's relatively small domestic backer pool limits how much capital a purely domestic crowdfunding campaign can realistically raise compared to larger crowdfunding markets (the UK or US), and reward-based crowdfunding in particular works best for the kind of "scalable" consumer-facing products backers can understand and want a reward from — it is a poor fit for capital-intensive, technical, or B2B ventures where the funding need far exceeds what a reward-based campaign can plausibly raise, or where there simply isn't a consumer reward to offer.

Mark Allocation — Part (c) (34 marks)

Tier	Marks	What separates this tier
26-34	26-34	Correctly contrasts crowdfunding's aggregation/signal mechanism against VC's concentrated due-diligence mechanism AND both AON and KIA correctly defined AND one advantage and one limitation given, each specific to the Irish SME context rather than generic.
17-25	17-25	Correct AON/KIA definitions and a correctly identified advantage and limitation, but the information-asymmetry contrast with VC is missing or superficial.

9-16	9-16	AON/KIA definitions muddled or reversed, or only one of the advantage/limitation pair given.
0-8	0-8	No credible definitions given, and no meaningful advantage or limitation identified.

Question B2 Total: 100 Marks